Collective Robotics Mini Project

Collective Resource Selection Through Local Sampling and Distributed Decision Making

Collective Robotics: Master-AI Universität zu Lübeck SS26
submission deadline: 14.08.2026 23:59

1 Introduction

Many collective systems in nature are capable of selecting beneficial resources through decentralized interactions. Individual agents gather information locally, exchange information with neighbors, and collectively converge toward a common decision without the need for a centralized coordinator. In this project, you will investigate how a swarm of robots can collectively identify and select the best resource site among several alternatives using only local sensing and local communication. You will design, model, implement, and evaluate a swarm robotic system capable of distributed collective decision making. The project combines:

- Requirement analysis
- Biological inspiration
- Agent-based modeling
- Micro–macro analysis
- Numerical simulation
- Experimental evaluation
- Scientific reporting

2 Learning Objectives

After completing this project, you should be able to:

1. Analyze a collective robotics problem and derive system requirements.
2. Extract collective decision-making principles from biological systems.
3. Design microscopic behavioral rules for autonomous agents.
4. Derive macroscopic population-level models.
5. Implement and simulate both microscopic and macroscopic models.
6. Validate the consistency between micro and macro descriptions.
7. Evaluate collective performance under varying conditions.
8. Interpret emergent behavior using quantitative metrics.

3 Problem Description

A swarm of N robots operates in an environment containing several candidate resource sites.

Each resource site has:

- A quality value
- A spatial location

The robots do not initially know which site is best.

Each robot:

- Has no global knowledge
- Can sense only locally
- Can communicate only with nearby robots
- Must make decisions autonomously

The swarm should collectively identify and agree **upon the highest-quality resource site** through local interactions.

Examples of resource sites include Charging stations / Food Source / Work Area/ Data collection area

4 Biological Inspiration

You shall study a natural collective system that performs resource selection.

Possible examples include:

Ant Food Source Selection

Ants discover and exploit profitable food sources through local interactions.

Honeybee Foraging

Bees communicate profitable food locations through waggle dances.

Animal Group Decision Making

Animals often choose feeding locations through decentralized interactions.

Deliverable (D1)

A 2–3 page review including:

- Description of the biological process
- Visualization of Behaviors and features in the form of diagrams, charts, forms, etc.
- References to scientific literature (especially from the teaching materials)
- Identification of mechanisms transferable to robotics

5 Requirement Analysis

Translate the biological system into engineering requirements.

5.1 Functional Requirements

The swarm shall:

- Explore the environment.
- Detect candidate resource sites.
- Estimate resource quality.
- Exchange information locally.
- Form opinions regarding resource preference.
- Reach consensus regarding the preferred resource.

5.2 Non-Functional Requirements

The swarm shall be:

- Scalable AND flexible
- Decentralized
- Fault tolerant

5.3 Assumptions

You must explicitly state assumptions regarding:

- Sensing
- Localization
- Communication
- Environment dynamics

6 Microscopic Model Design

Develop an agent-based model describing the behavior of individual robots.

6.1 Robot States

Each robot may occupy one of the following states:

Undecided (U)

The robot has not yet selected a preferred resource.

Supporting Resource A (A)

The robot believes Resource A is the best option.

Supporting Resource B (B)

The robot believes Resource B is the best option.

Additional states may be introduced if more than two resource sites are considered.

6.2 State Transition Rules

Define rules governing:

- Resource Discovery
- Resource Evaluation
- Opinion Exchange
- Opinion Switching

Resource Discovery

Probability of discovering a resource.

Resource Evaluation

Probability of supporting a resource based on observed quality.

Opinion Exchange

Mechanism through which neighboring robots exchange opinions.

Opinion Switching

Conditions under which a robot changes its preference.

Deliverables (D2)

- State machine diagram
- State transition equations
- Parameter definitions

7 Macroscopic Model Development

Develop a population-level model describing the collective dynamics.

Let $x_A(t)$ represent the fraction of robots supporting Resource A.

Let $x_B(t)$ represent the fraction of robots supporting Resource B.

Let $x_U(t)$ represent the fraction of undecided robots.

where

$$x_U = 1 - x_A - x_B$$

The macroscopic model should describe:

- Resource discovery
- Opinion formation
- Social influence
- Consensus formation

A possible baseline model is

$$\frac{dx_A}{dt} = \alpha_A x_U + \rho_A x_A x_U - \gamma_A x_A$$

$$\frac{dx_B}{dt} = \alpha_B x_U + \rho_B x_B x_U - \gamma_B x_B$$

where

- α = discovery rate
- ρ = social influence rate
- γ = opinion abandonment rate

Required Analysis

Your is tasks:

- Explain assumptions
- Derive model equations
- Discuss stability qualitatively

8 Micro–Macro Mapping

Establish explicit correspondence between:

Microscopic Parameter	Macroscopic Parameter
Discovery probability	Discovery rate
Opinion adoption probability	Social influence rate
Opinion switching probability	Abandonment rate
Resource quality estimate	Preference strength

Table 1: Example micro–macro mapping.

Your task is to explain follwoings points:

- Why the mapping is valid
- Under what conditions it fails
- Effects of finite swarm size

9 Numerical Implementation

Implement both models.

9.1 Agent-Based Simulation

Recommended tools:

- Python
- MATLAB
- NetLogo

The simulation should include:

- Environment representation
- Robot motion
- Resource discovery
- Local communication
- Opinion exchange

9.2 Macroscopic Simulation

Implement numerical integration using:

- ODE Solvers

Deliverables (D3)

Simulation code and documentation.

10 Experimental Evaluation

Conduct systematic simulation studies.

10.1 Experiment : Swarm Size

$$N = \{20, 50, 100, 200\}$$

Measure for each N the following values and report them in a table :

- Consensus time
- Decision accuracy

11 Model Validation

Compare microscopic and macroscopic predictions. Evaluate Fraction supporting each resource. Discuss the results using charts and diagrams.

Deliverables (D4)

code and documentation for reproducible experiment pipeline

12 Final Report

final report should contain following sections:

1. Introduction
2. Biological Inspiration
3. Microscopic Model
4. Macroscopic Model
5. Micro–Macro Mapping
6. Numerical Implementation
7. Experimental Setup
8. Results
9. Validation
10. Discussion
11. Conclusions
12. References

Expected report length: **8–15 pages, excluding references**.

13 Assessment Rubric

Component	Weight
Biological Study	10%
Microscopic Model	20%
Macroscopic Model	20%
Micro–Macro Mapping	10%
Numerical Implementation	20%
Experimental Evaluation	15%
Discussion and Conclusions	5%

Table 2: Project grading rubric.

13.1 Final Submission

Submission File: Your final submission should contain followings:

1. Final Report (PDF + Source (Latex project or Document file)
2. Project code with comments in it as documentation
3. Experiment pipeline
4. Short video (10-15 Minutes) explaining your code and project. Show your ID/Student-ID and introduce yourself at the beginning of video.
5. (Optional) PowerPoint presentation

Submission Notes (Please read carefully)

- The given deadline is firm and any submission after the deadline will not be accepted. Organize your work so that you have a buffer before the deadline . Technical issues on last minute are not acceptable excuses.
- Ensure your implementation matches your written report. Make sure, that your code is reproducible. Make your experiments code deterministic under a fixed seed.
- Please submit a compressed version of your video. You can use FFMPEG to compress your video file, or alternatively upload it to YouTube as a private video. After YouTube has processed the video, you can download it again, which often results in an optimized compressed version. Please note that by using YouTube, you are sharing your data with Google. If you have privacy concerns, consider using a local compression tool such as FFMPEG instead.
- Codes, final report, and video are mandatory. Missing any of them leads to an overall rejection of the mini-project.
- You are not allowed to use Large Language Models (LLMs) to solve this task. Any use of LLMs will be considered academic misconduct which result in you being barred from taking the final exam.